

BOWEI (JARRED) REN

Irvine, CA | jarredr1@uci.edu

Portfolio: psweb-theta.vercel.app | GitHub: github.com/JarredRbw

EDUCATION

University of California, Irvine - Irvine, CA

Expected June 2029

B.S. in Computer Engineering

TECHNICAL & CREATIVE SKILLS

Programming: C++, Java, Python, JavaScript/TypeScript, PHP, SQL

Web & Data: React, Next.js, HTML/CSS, Tailwind CSS, PostgreSQL, MySQL, Supabase, OAuth, Git, Vercel

Systems & Hardware: PC assembly/upgrades, component installation, BIOS/UEFI configuration, Windows troubleshooting, macOS, Raspberry Pi/Linux environments, drivers and peripherals

Networking & Low-Voltage: small LAN/Wi-Fi deployment, router/access-point configuration, Ethernet cabling/endpoints, connectivity troubleshooting and fault isolation

Creative & AV: digital photography, image curation, microphone/audio setup, stage-lighting operation

Language: Mandarin Chinese (Native)

SELECTED PROJECTS

AI Email Assistant - Independent Full-Stack Project

- Built a full-stack email management application with Next.js, TypeScript, Supabase, PostgreSQL, and Gmail OAuth; implemented incremental synchronization, AI classification, priority scoring, task queues, and scheduling.
- Validated workflows on a 250-email dataset and maintained 29 passing application tests while iterating on product behavior and persistence.

BJEA Campus Forum - Founder & Full-Stack Developer

- Built an internal discussion forum using PHP, MySQL, and custom routing; designed data structures, UI pages, moderation tools, and small AI utilities for student discussion and anonymous feedback.

Computer Vision & Edge AI

- Developed a CNN-BiLSTM Chinese sign-language recognition model with 90% test accuracy; handled sequence-data cleaning, model design, tuning, evaluation, research writing, and final defense.
- Built a YOLOv8 person-detection pipeline for Raspberry Pi 5 with real-time camera inference plus CPU, memory, temperature, and FPS monitoring; integrated human detection into a robotics project.

Chrome Web Time Tracker

- Created a Manifest V3 Chrome extension using Tabs, Storage, content scripts, and a service worker to track per-site browsing time locally and summarize visits without uploading page data.

TECHNICAL & AV EXPERIENCE

PC Hardware, Networking & Low-Voltage - Independent Technical Work

- Assemble, upgrade, and troubleshoot desktop PCs; configure BIOS/UEFI, operating systems, drivers, cooling, storage, power delivery, and peripherals.
- Set up and troubleshoot small wired/wireless networks and basic low-voltage installations, including routers/access points, Ethernet runs, endpoint connectivity, and practical fault isolation.

Lighting & Sound Technician - School Drama and Talent Shows

- Managed rehearsals, microphone setup and audio adjustments, and designed/operated stage-lighting effects for Chinese/English drama festivals and campus talent shows.

RECOGNITION

ACSL Final Round - Top 30% | HiMCM - Honorable Mention